Orbits, Forces and Stability: Exploring an N-Body System

SimuPhysics N-Body Gravity Simulator · SimuPhysics

Years 7-10	~30 minutes	Science · Physics

NOTICE — AI-generated worksheet
This worksheet was drafted automatically from the site’s curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: https://simuphysics.com/view/mechanics-n-body-gravity-simulator.html
Curriculum links: AC9S7U04 · AC9S10U05

Learning intentions

- I can describe gravitational force as a force acting between two masses.
- I can set up a stable two-body orbit and explain what keeps it stable.
- I can predict and explain how adding a third mass changes the stability of an orbiting system.

Getting started

Open the simulator and set up a simple two-body system first — a large central mass with a smaller mass given an initial sideways velocity — and let it run until you see a stable orbit form. Keep this two-body setup running while you complete the first few tasks, then add a third mass for the later ones.

Tasks

1. Describe the two-body orbit you created. Is the smaller mass's path a circle, an ellipse, or something else?

2. Increase the initial velocity of the smaller mass. Describe what happens to the orbit.

3. Explain, in your own words, why the smaller mass doesn't simply fall straight into the larger one, using the idea of balanced and unbalanced force.

4. Increase the mass of the central body while keeping everything else the same. What happens to the force acting on the orbiting mass?

5. Add a third mass in three different starting positions and record the outcome for the system each time.

Position of third mass	What happened to the system

6. Based on your table, explain what generally happens to a stable two-body system when a third mass is introduced nearby.

7. Predict what would happen if the third mass were much larger than the other two. Test your prediction in the simulation and describe the result.

Extension (optional)

Gravitational force depends on both mass and the inverse square of the distance between objects. Use this to explain why a mass placed very close to the central body behaves very differently to one placed far away, even if both masses are the same size.